COMMUNICATION MODULE

National Freelance & Skill Verification Platform

SUB-PROJECT MANAGEMENT PLAN

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1. EXECUTIVE SUMMARY

The Communication Module is a sub-project of the National Freelance and Skill Verification Platform. Its purpose is to provide secure, real-time communication between freelancers and clients, enabling efficient collaboration throughout project lifecycles.

The module includes real-time messaging, file sharing, notifications, meeting link integration (Zoom/Google Meet), group chats, and message history. The goal is to ensure seamless, transparent, and reliable communication within the platform.

1.1 Project Goals

- To establish a centralized communication system within the platform
- To enhance collaboration between freelancers and clients
- To reduce dependency on external communication tools
- To improve overall user experience and satisfaction

1.2 Project Objectives

- Develop a real-time messaging system with minimal latency
- Enable secure file sharing within conversations
- Implement a reliable notification system for user interactions
- Provide support for group chats to facilitate team collaboration
- Integrate third-party meeting platforms (Zoom/Google Meet)
- Ensure secure storage and retrieval of message history
- Design a user-friendly and responsive communication interface
- Achieve successful integration with other platform modules

2. PROJECT MANAGEMENT APPROACH AND GOVERNANCE

The Communication Module will follow an **Agile-based development approach**, progressing iteratively through planning, design, development, testing, and integration phases. This enables flexibility, continuous feedback, and incremental delivery while aligning with the defined schedule baseline. Regular team meetings and coordination with the Integration Group will ensure timely progress, dependency management, and early risk identification.

2.0.1 Roles and Responsibilities

The project team consists of four members:

- **Team Lead:** Planning, task allocation, stakeholder coordination, progress monitoring, and reporting
- **Backend Developer:** Server-side logic, messaging system, APIs, and database management
- **Frontend Developer:** User interface, chat features, and user interaction design
- **QA & Integration Engineer:** Testing, debugging, validation, and integration support

2.0.2 External Resources

- Third-party APIs, Zoom and Google Meet, for meeting link integration
- Development and tracking tools like GitHub.

2.0.3 Assumptions

- Third-party APIs will be stable and accessible
- Required modules (e.g., authentication, profiles) will be available for integration
- Team members have the required skills and will adhere to the project schedule

2.0.4 Constraints

- Fixed academic timeline (schedule constraint)
- Dependency on other modules for full functionality (integration risk)
- Limited resources and no dedicated budget for external services

2.1 PROJECT SCOPE

Scope Statement

Sub-Project Title: Communication Module – National Freelance and Skill Verification Platform
Date: 25, March 2026
Prepared by: Haifa

2.1.1 Project Summary and Justification

The Communication Module is a core component of the National Freelance and Skill Verification Platform. This project is initiated to ensure secure, efficient, and real-time communication between freelancers and clients, which is essential for successful project execution and collaboration.

Currently, communication gaps, delays, and lack of centralized interaction tools reduce productivity and transparency in freelance environments. This module aims to resolve these issues by providing an integrated communication system within the platform.

The module will improve:

- Collaboration efficiency between users
- Transparency in project discussions
- Overall user satisfaction and platform reliability

2.1.2 Product Characteristics and Requirements

1. **Real-time Messaging:**
   The system will allow users (freelancers and clients) to send and receive messages instantly using a real-time one-to-one communication mechanism.
2. **File Sharing:**
   Users will be able to upload, send, and download files (documents, images, etc.) within chat conversations. Files must be securely stored and accessible.
3. **Message History:**
   All conversations will be stored and retrievable. Users will be able to view past messages and maintain communication records.
4. **Message Status Indicator:**
   All messages sent will have a status of one of the following: pending, sent, received.
5. **Notifications:**
   The system will provide real-time notifications for new messages, and group activity. Notifications will be timely and reliable.
6. **Group Chats:**
   The module will support group communication between multiple freelancers and clients working on the same project. Users must be able to create and manage group chats.
7. **Voice/Video Meeting Links:**
   The system will allow users to generate and share meeting links using third-party platforms such as Zoom or Google Meet. No custom video system will be developed.

8. **Security/End-to-End Encryption:**
   The communication system must ensure secure data transmission and storage. Only authorized users should have access to conversations. Proper authentication and access control must be enforced.
9. **System Availability:**
   The communication module must be available 24/7 with minimal downtime, except for scheduled maintenance.
10. **User Interface:**
    The chat interface must be user-friendly, responsive, and accessible through supported browsers and devices.

2.1.3 Summary of Project Deliverables

2.1.3.1 Process Deliverables

- Sub-Project Management Plan (SPMP)
- Requirements Management Plan (SP-RMP)
- Software Requirements Specification (SP-SRS)
- Software Design Specification (SP-SDS)
- Project charter
- Scope statement
- Work Breakdown Structure (WBS)
- Schedule baseline
- Milestones list
- Progress and status reports
- Risk management documentation
- Final project report

2.1.3.2 Product Deliverables

1. **Messaging System:**
   Fully functional real-time messaging feature for one-to-one communication.
2. **Group Chat System:**
   Ability to create and manage group conversations for project teams.
3. **File Sharing System:**
   Secure upload, storage, and sharing of files within chats.
4. **Notification System:**
   Real-time alerts for messages, files, and updates.
5. **Meeting Integration Feature:**
   Functionality to generate and share Zoom/Google Meet links.
6. **Message History Database:**
   Storage and retrieval system for all chat data.
7. **User Interface Design:**
   Chat screens, notification panels, and interaction flows.
8. **Test Plan:**
   Documentation for testing processes, test cases, and bug reporting.

9. **Integration Support:**
Ensuring compatibility with other modules like user management and project management.

2.1.4 Project Success Criteria

The Communication Module will be considered successful if:

- All core features (messaging, file sharing, notifications, group chats, meeting links) are fully functional
- The system provides real-time communication with minimal delay
- Communication is secure and accessible only to authorized users
- The module integrates successfully with the overall system
- The project is completed within the defined academic timeline
- Users can effectively communicate without external tools

Even if minor delays occur, the project will be considered successful if it delivers a **reliable, scalable, and user-friendly communication system** that significantly enhances collaboration on the platform.

2.2 DELIVERABLES

Refer to part 2.1.3 of the scope statement under 2.1 Project Scope

2.3 WORK BREAKDOWN STRUCTURE (WBS)

2.3.1 How WBS Will Be Used

The Work Breakdown Structure (WBS) decomposes the Communication Module into manageable work packages to ensure systematic planning, execution, and tracking of project activities. It provides a hierarchical structure that aligns project scope with deliverables and supports effective schedule and resource management.

The WBS for this module is organized into major phases including planning, design, development, testing, deployment, and documentation. Each phase is further divided into specific tasks to ensure clarity of responsibilities and progress monitoring.

2.3.2 Deliverable-Based Tabular WBS

6.0 Communication Module

- **6.1 Project Management Deliverables**
  - 6.1.1 Project Plan
  - 6.1.2 Meeting Records & Coordination Logs
  - 6.1.3 Progress Reports
- **6.2 Requirements Deliverables**
  - 6.2.1 Requirements Management Plan (SP-RMP)
  - 6.2.2 Requirements Specification (SP-SRS)
  - 6.2.3 Requirements Analysis Report
- **6.3 System Design Deliverables**
  - 6.3.1 System Architecture
  - 6.3.2 Database Schema (Messages, Files, Chat History)
  - 6.3.3 API Design Specification
  - 6.3.4 UI/UX Design Mockups
- **6.4 Messaging System Deliverables**
  - 6.4.1 Real-time Messaging System
  - 6.4.2 Message Delivery & Synchronization
- **6.5 Group Chat System Deliverables**
  - 6.5.1 Group Creation & Management
  - 6.5.2 Multi-user Messaging
- **6.6 File Sharing System Deliverables**
  - 6.6.1 File Upload & Sharing
  - 6.6.2 Secure File Storage
- **6.7 Notification System Deliverables**
  - 6.7.1 Real-time Notification System
  - 6.7.2 Notification Trigger Mechanism
- **6.8 Meeting Integration Deliverables**
  - 6.8.1 Meeting Link Generation
  - 6.8.2 Meeting Link Sharing
- **6.9 Message History & Database Deliverables**
  - 6.9.1 Message Storage System
  - 6.9.2 Chat History Retrieval
- **6.10 User Interface Deliverables**
  - 6.10.1 Chat Interface Screens
  - 6.10.2 Notification Panels
  - 6.10.3 User Interaction Flows
- **6.11 Integration Deliverables**
  - 6.11.1 Messaging API Services

- 6.11.2 Integration with User Module
- 6.11.3 Integration with Project/Collaboration Modules

- **6.12 Testing Deliverables**
  - 6.12.1 Unit Test Results
  - 6.12.2 Integration Test Results
  - 6.12.3 System Test Reports
  - 6.12.4 Bug Reports & Fix Logs
- **6.13 Documentation Deliverables**
  - 6.13.1 SP-SRS Document
  - 6.13.2 SP-SDS Document
  - 6.13.3 Technical Documentation
- **6.14 Deployment & Final Deliverables**
  - 6.14.1 Deployed Communication Module
  - 6.14.2 Final Demonstration
  - 6.14.3 Final Report

2.3.3 WBS Dictionary

WBS ID	Task Name	Description	Duration	Cost	Resources (# & Type)
6.1.1	Project Plan	Defines scope, schedule, and execution strategy	1 day	0	1 (Team Lead)
6.1.2	Meeting Records	Records meetings and coordination	Continuous	0	1 (Team Lead)
6.1.3	Progress Reports	Tracks module progress	Continuous	0	2 (Team Lead + QA)
6.2.1	SP-SRS	Functional & non-functional requirements	2 days	0	4 (All Team)
6.2.2	Requirements Analysis	Analyze and validate requirements	1 day	0	2 (Team Lead + Dev)
6.2.3	Requirements Documentation	Document finalized requirements	1 day	0	2 (Team Lead + Dev)
6.3.1	System Architecture	Overall system design and flow	3 days	0	2 (Backend + Frontend)

6.3.2	Database Schema	Design for messages, files, history	2 days	0	1 (Backend Dev)
6.3.3	API Design	Define APIs for communication	2 days	0	1 (Backend Dev)
6.3.4	UI/UX Design	Design interfaces and flows	2 days	0	1 (Frontend Dev)
6.4.1	Messaging System	Real-time one-to-one messaging	3 days	0	2 (Backend + Frontend)
6.4.2	Message Sync	Message delivery & synchronization	1 day	0	1 (Backend Dev)
6.5.1	Group Management	Create and manage groups	1 day	0	1 (Backend Dev)
6.5.2	Group Messaging	Multi-user chat system	2 days	0	2 (Backend + Frontend)
6.6.1	File Sharing	Upload and share files	2 days	0	2 (Backend + Frontend)
6.6.2	Secure Storage	Secure storage of files	1 day	0	1 (Backend Dev)
6.7.1	Notification System	Real-time alerts	1 day	0	1 (Backend Dev)
6.7.2	Notification Trigger	Alert generation mechanism	1 day	0	1 (Backend Dev)
6.8.1	Link Generation	Generate meeting links	1 day	0	1 (Backend Dev)
6.8.2	Link Sharing	Share links in chat	1 day	0	1 (Backend Dev)
6.9.1	Message Storage	Store chat data	2 days	0	1 (Backend Dev)
6.9.2	Chat Retrieval	Retrieve chat history	1 day	0	1 (Backend Dev)
6.10.1	Chat UI	Messaging interface	2 days	0	1 (Frontend Dev)

6.10.2	Notification UI	Alerts interface	1 day	0	1 (Frontend Dev)
6.10.3	Interaction Flows	User navigation design	1 day	0	1 (Frontend Dev)
6.11.1	Messaging APIs	Backend communication services	2 days	0	1 (Backend Dev)
6.11.2	User Integration	Integration with user module	1 day	0	1 (Backend Dev)
6.11.3	Module Integration	Integration with other modules	2 days	0	2 (Backend + QA)
6.12.1	Unit Testing	Test individual components	1 day	0	1 (QA)
6.12.2	Integration Testing	Test module interactions	2 days	0	2 (QA + Backend)
6.12.3	System Testing	End-to-end testing	2 days	0	1 (QA)
6.12.4	Bug Logs	Record and fix issues	Continuous	0	2 (QA + Dev)
6.13.1	SP-SRS Doc	Final requirements document	2 days	0	4 (All Team)
6.13.2	SP-SDS Doc	Final design document	2 days	0	2 (Dev Team)
6.13.3	Technical Docs	Implementation documentation	2 days	0	2 (Dev Team)
6.14.1	Deployment	Deploy module	1 day	0	2 (Backend + QA)
6.14.2	Demonstratio n	Final presentation	1 day	0	4 (All Team)
6.14.3	Final Report	Complete documentation	2 days	0	4 (All Team)

2.4 STAKEHOLDER ANALYSIS

Stakeholders for this module were identified based on their involvement in the project and the impact the project will have on their roles. The identification process included analyzing users, development contributors, system integrators, and oversight authorities. Each stakeholder has a specific interest in the project and varying levels of influence on decision-making.

2.4.1 Stakeholder Identification

- **Freelancers & Clients:**
  Primary users of the module. They rely on the system for real-time communication and collaboration. Their feedback will directly impact feature usability and system enhancements.
- **Module Team:**
  Responsible for designing, coding, and implementing the module. Their work determines the functionality, performance, and integration of the system.
- **Integration Group:**
  Ensures the module integrates seamlessly with other platform components. They impact deployment decisions and system compatibility.
- **Project Instructor:**
  Provides oversight, approval, and academic guidance. Their evaluation impacts project success and deliverables quality.

2.4.2 Stakeholder Analysis Matrix

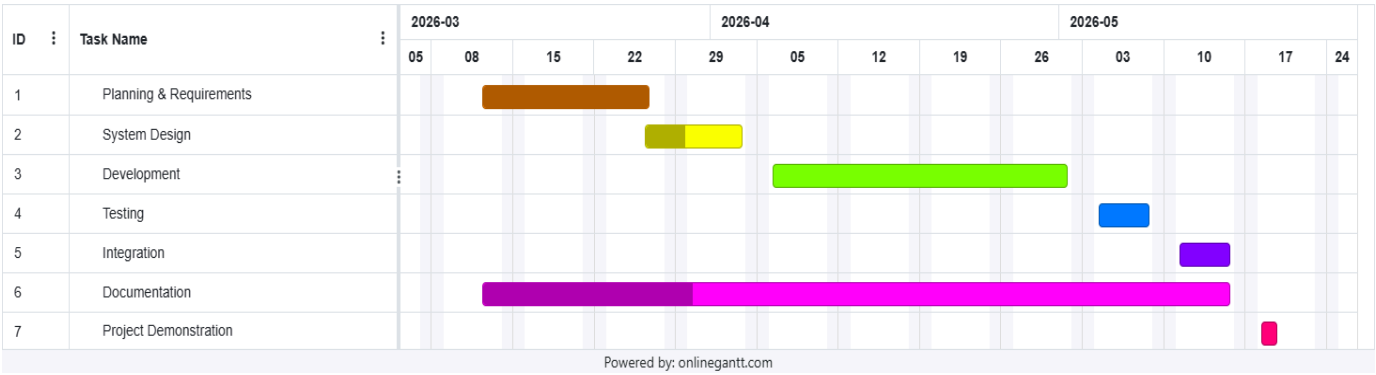
Stakeholder	Role	Interest	Influence	Impact of Project	Management Strategy
Freelancers & Clients	End users	High	Medium	Improved collaboration and communication	Gather feedback, usability testing, regular updates
Module Team	Implementation	High	High	Responsible for successful development	Agile iterations, regular meetings, progress tracking
Integration Group	System integration	High	High	Ensures compatibility with other modules	Coordinate integration schedule, risk identification
Project Instructor	Oversight	High	High	Approves deliverables, guides project	Regular status reports, consultation on decisions

2.5 SCHEDULE BASELINE

2.5.1 Phase Schedule

Phase	Duration	Description
Planning	1 week	Requirements gathering, stakeholder analysis, documentation
Design	1 week	System architecture, database design, API integration planning
Development	4 weeks	Implementation of messaging, file sharing, notifications, meeting link integration, and group chat
Testing	1 week	Unit testing, integration testing, and performance testing
Integration	1 week	Integration with the main platform and final validation
Documentation	Parallel	Preparation of RMP, SRS, SDS throughout all phases

2.5.2 Gantt Chart



2.6 MILESTONE LIST

2.6.1 Summary

Our communications module includes the following major milestones to ensure progress tracking and timely delivery. Each milestone represents a critical completion point of a phase or key deliverable.

1. **Requirements & Stakeholder Approval** – Completion of all requirements and stakeholder analysis.
2. **System Design Complete** – Architecture, database schema, and API design finalized.
3. **Development Complete** – All core features (messaging, file sharing, notifications, meeting links, group chat) implemented.
4. **Testing Complete** – Unit, integration, and performance testing finalized.
5. **Integration Complete** – Module integrated with main platform and validated.
6. **Documentation Complete** – RMP, SRS, SDS finalized.
7. **Project Demonstration** – Instructor review and final demonstration of completed module.

2.6.2 Schedule Update and Change Management

1. **Monitoring & Reporting**
   - Weekly team meetings will review progress against milestones.
   - Status reports will highlight any delays or risks.
2. **Updating Schedule & Milestones**
   - Any proposed changes to the schedule or milestones must be documented, justified, and submitted to the Team Lead.
   - Changes require approval from the project team and, if necessary, the instructor.
3. **Communication of Changes**
   - Approved schedule changes will be communicated via email and updated in the project tracking tool (e.g., GitHub, shared calendar).
   - Team members, integration group, and instructor will be notified of revised milestone dates and impacts.

2.6.3 Milestones Table

MILESTONE	DESCRIPTION	DATE
M1: Requirements & Stakeholder Approval	Completion of requirements gathering, stakeholder analysis, and approval of scope	Week 1
M2: System Design Complete	Architecture, database, and API design finalized and approved	Week 2
M3: Development Complete	All features implemented (messaging, file sharing, notifications, meeting links, group chat)	Week 6
M4: Testing Complete	Unit, integration, and performance testing completed	Week 7

M5: Integration Complete	Module integrated with main system and validated	Week 7
M6: Documentation Complete	RMP, SRS, SDS finalized and approved	Parallel throughout; complete by Week 8
M7: Project Demonstration	Final presentation and demonstration to instructor	Week 8

2.7 CHANGE MANAGEMENT PLAN

1. **Submission of Change Requests**
    - Any stakeholder, including team members, freelancers, clients, integration group, or the instructor, may submit a change request.
    - Requests must include a description of the change, justification, and potential impact on scope, schedule, and resources.
2. **Review of Changes**
    - The **team lead** is responsible for the initial review of all change requests, assessing feasibility, risks, and resource impact.
    - The **integration group** coordinates to ensure the change does not conflict with other modules or dependencies.
3. **Approval of Changes**
    - Minor changes may be approved by the **team lead** with integration oversight.
    - Major changes require **team consensus** and, if necessary, **instructor approval.**
    - All approved changes are incorporated into the project schedule, WBS, and documentation.
4. **Documentation and Tracking**
    - Every change request will be logged in a **Change log**, including submission date, description, impact analysis, approval status, and implementation details.
    - Version control will be maintained for all project documents (RMP, SRS, SDS, codebase) to reflect approved changes.
5. **Communication of Changes**
    - Approved changes will be communicated to all stakeholders via email and project tracking tools (e.g., GitHub, shared calendar).
    - The change log and any updated documentation will be accessible to the project team and integration group for reference.

2.8 PROJECT SCOPE MANAGEMENT PLAN

2.8.1 Scope Definition and Authority

- The scope of the communication module is defined during the **kickoff and planning phases**.
- The **team lead** has the **authority** to manage and monitor the project scope on behalf of the group.
- Any adjustments to scope require formal approval before implementation.

2.8.2 Scope Measurement

- Scope will be measured by comparing **planned deliverables** against the **Software Requirements Specification (SRS)** and **Work Breakdown Structure (WBS)** tasks.
- Completion of each WBS task, milestone, and feature (e.g., messaging system, file sharing, group chat) will be used to validate that the module scope is being met.
- Deliverables will be reviewed weekly in team meetings to identify deviations and ensure alignment with the defined scope.

2.8.3 Scope Change Process

- Proposed changes to scope must be formally documented in a **Change Request Form** and submitted to the Team Lead.
- Minor changes may be approved by the Team Lead with consultation from the Integration Group.
- Major changes require approval from the **integration group** and, if necessary, the instructor.
- All changes will be tracked, logged, and communicated to all stakeholders, and relevant documents (SRS, WBS, SPMP) will be updated accordingly.

2.8.4 Final Scope Approval

- Deliverables will be **validated via SRS** and approved upon successful testing and verification.
- **Final approval** of the module scope will be provided jointly by the **leader** and the **integration group** to ensure compliance with overall system integration requirements.

3. COMMUNICATION MANAGEMENT PLAN

3.1 Stakeholder Contact Information

NAME	TITLE	EMAIL	PERSONAL PHONE
Amna Shazad	Team Lead	i233073@isb.nu.edu.pk	+923340092888
Haifa Yousaf	Developer	i233072@isb.nu.edu.pk	+923347221155
Tanees Fatima	Developer	i233003@isb.nu.edu.pk	+923004000006
Nisha Zafran	QA & Integration Engineer	i233023@isb.nu.edu.pk	+923165897957
Freelancers & Clients	End Users	N/A	N/A
Abdul Basit (Integration Group Leader)	System + Project Integration	i233018@isb.nu.edu.pk	N/A
Dr. Behjat Zuhaira(Project Instructor)	Oversight	behjat.zuhaira@nu.edu.pk	N/A

3.2 Communication Plan

COMMUNICATION TYPE	DESCRIPTION	FREQUENCY	MESSAGE DISTRIBUTION	DELIVERABLE	DELIVERABLE OWNER
Team Meetings	Discuss progress, task updates, and blockers	Daily	Online (Zoom/Google Meet)	Meeting notes	Team Lead
Integration Meetings	Align module with other sub-projects	Weekly	Online	Integration status report	QA & Integration group
Status Reports	Weekly progress summary, milestones, and risks	Weekly	Assignments / Document	Progress report	Team Lead
Issue Tracking	Record bugs, issues, and resolutions	Continuous	GitHub/Jira	Issue log	QA & Integration group
Client Feedback	Gather input from freelancers/clients on module functionality	After milestone completion	Online forms / Email	Feedback report	Team Lead
Instructor Updates	Inform instructor of project status and deliverables	Milestone completion	Assignments / Document	Progress updates	Team Lead & Integration group
Ad-hoc Communications	Immediate alerts for urgent issues	As needed	Email / Messaging	Action items	Responsible Team Member

3.3 Communication Leaders List

NAME	ROLE
Amna Shahzad	Module 6 Lead
Ayesha Naveed	Module 1 Lead
Muhammad Hashir	Module 2 lead
Immad Babar	Module 3 lead
Inam Ullah	Module 4 lead
Rizwan Saeed	Module 5 lead
Taha Salam	Module 7 lead
Sultan	Module 8 lead
Salman Ahmed	Module 9 lead
Ubaidullah	Module10 lead
Manahil	Module 11 lead
Muhammad Ibrahim	Module 12 lead
Moeez Abid, Ahnaf Abdullah, Noor Alam, Abdul Basit	Integration Group

4. RESOURCE MANAGEMENT PLAN

4.1 Resource Categories

1. **Development Tools**
   - GitHub: For version control, issue tracking, and collaboration.
   - VS Code: Integrated development environment for coding frontend and backend.
   - **Procurement:** No purchase required. Free and academic licenses will be used.
2. **Backend Services**
   - Firebase / WebSockets: For real-time messaging, database, and notifications.
   - **Procurement:** Free/educational plans will be used; no physical equipment required.
3. **Third-Party APIs**
   - Zoom / Google Meet API: For meeting link generation and integration.
   - **Procurement:** Free-tier APIs will be used; no hardware required.
4. **Infrastructure**
   - Cloud backend hosting: Supports database, messaging system, and notifications.
   - **Procurement:** Through cloud services (no physical servers).

4.2 Resource Allocation

Resource	Type	Assigned To	Notes
GitHub	Tool - Free	All Team Members	Version control, issue tracking
VS Code	Tool - Free	Frontend & Backend Developers	Coding environment
Firebase / WebSockets	Backend Service - Free	Backend Developer	Real-time messaging & database
Zoom / Google Meet API	API Service - Free	Backend Developer	Meeting link integration
Cloud Backend	Infrastructure - Free/Cloud	Backend Developer	Hosting & storage

4.3 Procurement and Management

- All resources are **virtual or cloud-based**, requiring no physical procurement.
- Services and tools will be accessed using **team or academic licenses**.
- The Team Lead is responsible for ensuring all team members have access and required permissions.
- Usage and allocation will be monitored during weekly meetings and recorded in project logs.

5. HUMAN RESOURCES MANAGEMENT PLAN

Staffing needs for the Communication module project will be determined based on the scope of work, technical requirements, and project schedule. Since the project is divided into phases such as planning, design, development, testing, and integration, human resources will be allocated accordingly to ensure efficiency and timely completion.

The project team consists of four members, each assigned a specific role based on their skill set. Required skill sets include backend development (real-time messaging systems, APIs, database handling), frontend development (user interface design and chat systems), quality assurance (testing and debugging), and project coordination.

As this is an academic project, no salaries or hourly rates are applicable. All team members contribute equally as part of coursework requirements.

Training requirements are minimal; however, team members may need to familiarize themselves with tools and technologies such as WebSockets/Firebase for real-time communication, API integration (Zoom/Google Meet), and version control systems like GitHub.

Once roles are finalized, each member will be assigned responsibilities and will be accountable for deliverables within their domain.

5.1 SUB-PROJECT STAFF LIST

NAME	TITLE	EMAIL	PHONE	TASKS ASSIGNED
Amna Shahzad	Team Lead	i233073@isb.nu.edu.pk	0334 0092888	• **6.1.1** Project Plan • **6.2.2** Requirements Analysis Report • **6.1.2** Requirements Management Plan (SP-RMP) • **6.1.3** Progress & Status Reports • **6.1.8** Risk Management Documentation • **6.14.2** Final Demonstration • **6.14.1** Final Validation (Deployment Oversight) • **6.13** Documentation Review & Compilation
Haifa Yousaf	Developer	i233072@isb.nu.edu.pk	0334 7221155	• **6.3.1** System Architecture • **6.3.2** Database Schema (Messages, Files, Chat History) • **6.3.3** API Design Specification • **6.6.1** File Sharing System • **6.9.1** Message Storage System • **6.13.2** Software Design Specification (SP-SDS) • **6.13.3** Technical Documentation
Tanees Fatima	Developer	i233003@isb.nu.edu.pk	0300 4000006	• **6.4.1** Real-time Messaging System • **6.5.2** Group Chat System • **6.7.1** Notification System • **6.8.1** Meeting Integration Feature • **6.4.2** Message Synchronization Mechanism • **6.10.1** Chat Interface Screens • **6.10.2** Notification Panels • **6.13.2** Software Design Specification (SP-SDS)

Nisha Zafran	QA and Integration Engineer	i233023@isb.nu.edu.pk	0316 5897957	• **6.12.1** Unit Test Results • **6.12.2** Integration Test Results • **6.12.3** System Test Reports • **6.12.4** Bug Reports & Fix Logs • **6.11.3** Integration with Other Modules • **6.2.1** Software Requirements Specification (SP-SRS) • **6.12** Test Documentation

5.2 RESOURCE REQUIREMENT CALENDAR

ROLE	HOURS PER MONTH			
	MAR	APR	MAY	JUN
Team Lead	40 hrs	20 hrs	10 hrs	-
Backend Developer	60 hrs	40 hrs	20 hrs	-
Frontend Developer	60 hrs	40 hrs	20 hrs	-
QA and Integration Engineer	20 hrs	40 hrs	30 hrs	-

6. SCHEDULE MANAGEMENT PLAN

6.1 Overview

The Schedule Management Plan defines the approach and processes that will be used to develop, monitor, and control the project schedule for the Communication Module. It ensures that all project activities are completed within the defined timeline and aligned with project milestones.

6.2 Schedule Development

The project schedule will be developed based on the Work Breakdown Structure (WBS), where tasks are divided into phases including planning, design, development, testing, and integration. Each task will be assigned a duration and dependencies to ensure proper sequencing of activities.

A Gantt chart will be used to visually represent the schedule and track task progress over time (shown above in schedule baseline).

6.3 Schedule Monitoring and Control

Project progress will be monitored through regular team meetings and updates. Tools such as Trello or Jira will be used to track task completion and identify delays. Any deviations from the planned schedule will be analyzed and corrective actions will be taken to minimize impact.

6.4 Schedule Changes Management

Any changes to the project schedule will be documented and approved by the Team Lead. Adjustments will be made in coordination with team members to ensure that deadlines are met without compromising quality.

7. QUALITY MANAGEMENT PLAN

7.1 Overview

The Quality Management Plan defines the processes, standards, and activities that will be followed to ensure that the Communication Module meets the required quality benchmarks in terms of functionality, performance, reliability, and user experience.

7.2 Quality Objectives

The primary objective is to ensure that the Communication Module delivers secure, real-time, and reliable communication between users without delays, failures, or data loss. The system must maintain high usability standards while ensuring data privacy and system stability.

7.3 Quality Assurance Processes

Quality will be ensured through a combination of preventive and corrective processes throughout the development lifecycle. During the design phase, architecture reviews will be conducted to ensure scalability and modularity. During development, coding standards and peer reviews will be enforced to maintain consistency and reduce defects.

- Testing will play a central role in quality assurance. This includes:
- Unit testing for individual components (e.g., messaging API)
- Integration testing between modules (e.g., communication + notification)
- System testing for complete workflows (e.g., sending messages with attachments)
- User acceptance testing (UAT) to validate real-world usability

7.4 Quality Standards

- The module will adhere to the following standards:
- IEEE SRS Standards for requirement clarity
- Secure coding practices (OWASP guidelines)
- Performance benchmarks (real-time messaging under 2 seconds)
- UI/UX standards for usability and accessibility

7.5 Continuous Improvement Process

Quality improvement will be an ongoing process. Feedback will be collected from testing phases and users, and iterative improvements will be made. Bugs and performance issues will be tracked, analyzed, and resolved systematically. Sprint reviews and retrospectives (if Agile is followed) will be used to identify areas for improvement and optimize development practices.

7.6 Quality Governance

- A structured governance model will be followed:
- Project Manager: Oversees quality compliance
- QA Team: Responsible for testing and validation
- Developers: Responsible for code quality and fixes
- Stakeholders: Provide feedback during reviews

- Regular checkpoints will ensure that quality standards are consistently met.

7.7 Quality Metrics

The following metrics will be used to evaluate quality:

- Message delivery time (< 2 seconds)
- System uptime (> 99%)
- Bug density (number of defects per module)
- User satisfaction (feedback ratings)
- Error rate in message delivery

7.8 Reporting Frequency and Methods

Quality reports will be generated weekly and shared with the project team. These reports will include:

- Test results
- Defect reports
- Performance analysis

Additionally, milestone-based reviews will be conducted at the end of each major development phase.

8. RISK MANAGEMENT PLAN

8.1 Overview

The Risk Management Plan outlines the approach for identifying, analyzing, and managing risks associated with the Communication Module to ensure smooth project execution.

8.2 Risk Identification

Risks will be identified throughout the project lifecycle using:

- Brainstorming sessions
- Past project analysis
- Expert judgment
- Technical reviews

Potential risks include technical failures, integration issues, security vulnerabilities, and delays.

8.3 Risk Analysis and Prioritization

Each identified risk will be analyzed based on:

- Likelihood of occurrence (Low, Medium, High)
- Impact on the project (Low, Medium, High)

A risk priority level will then be assigned, allowing the team to focus on high-risk issues first.

8.4 Risk Tracking and Monitoring

All risks will be documented in a Risk Log, which will be continuously updated. The project team will review risks on a weekly basis to track their status and ensure timely mitigation.
A risk owner will be assigned to each major risk, responsible for monitoring and managing it.

8.5 Risk Mitigation Strategies

Preventive measures will be applied to reduce the likelihood of risks. For example:

- Using proven technologies to reduce technical risk
- Conducting early integration testing
- Implementing secure authentication and encryption

8.6 Contingency Plans

If risks materialize, contingency plans will be executed. Examples include:

- Switching to alternative APIs if third-party services fail
- Implementing backup servers in case of system failure
- Allocating extra time for resolving integration issues

These plans ensure that this project can continue with minimal disruption.

8.7 RISK LOG

Risk ID	Description	Probability	Impact	Priority	Mitigation Strategy	Status
R1	Delay in real-time messaging implementation	Medium	High	High	Use WebSocket libraries and early testing	Open
R2	Integration failure with notification module	Medium	Medium	Medium	Conduct integration testing early	Open
R3	Security vulnerability in messaging system	Low	High	High	Implement encryption and security audits	Open
R4	Third-party API failure (Zoom/Meet)	Medium	Medium	Medium	Provide alternative API options	Open
R5	High server load causing delays	Medium	High	High	Use scalable cloud infrastructure	Open
R6	Data loss due to system crash	Low	High	High	Implement backups and recovery systems	Open

9. COST BASELINE

PROJECT PHASE	BUDGETED TOTAL (PKR)	COMMENTS
Requirements & Planning	0	No cost; performed by students as part of coursework
Design Phase	0	System design and architecture done by students
Development Phase	0	No costs for APIs, libraries, or development tools when using free packages.
Testing Phase	0	manual testing or free tool used
Deployment Phase	0	Basic cloud hosting or local hosting free of cost
Communication APIs	0	Use of free tiers (e.g., WebSockets, limited API usage)
Version Control & Tools	0	GitHub free tier / student access
Total	**=0 PKR**	

10. QUALITY BASELINE

ITEM	ACCEPTABLE LEVEL	COMMENTS
Message Delivery Time	≤ 2 seconds	Must meet SRS performance requirement
System Availability	≥ 99% uptime	Should remain accessible during usage
Message Accuracy	100% correctness	Messages must not be altered or lost
Bug Density	<=5 minor bugs/module	no critical bug allowed
Security (Encryption)	mandatory	HTTPS & secure communication required
File Upload Validation	100% validation	Only allowed file types accepted
Notification Delivery Rate	≥ 95% success	Notifications should reliably reach users

UI Responsiveness	≤ 3 seconds load time	Smooth user experience
Compatibility	major browsers supported	Chrome,Edge
Requirement Traceability	100% coverage	All requirements must map to test cases

11. APPENDICES

11.1 Glossary

TERM	DEFINATION
API	application Programming Interface used for communication between software components
UI	User Interface through which users interact with the system
Backend	Server-side logic handling data processing and communication
Real-time Messaging	Instant message delivery without noticeable delay
Encryption	Process of securing messages so only authorized users can read them
WebSocket	Protocol enabling full-duplex communication channels over a single TCP connection
Module	A specific functional component of the overall system

11.2 Acronyms

ACRONMYS	FULL FORM
SPMP	Software Project Management Plan
SRS	Software Requirements Specification
UI	User Interface
DB	Database
TC	Test Case
API	Application Programming Interface

11.3 References

- IEEE Std 1058-1998, Software Project Management Plans
- Course book

12. AUTHORIZATION SIGNATURES

PREPARED BY

Amna Shahzad (Group Leader), Haifa Yousaf, Nisha Zafran, Tanees Fatima

Names

29/03/2026

Group Leader Signature *Date*

APPROVED BY

Integration Group Leader

Signature *Date*